

# Final Report:

## AI Job Salary Analysis

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### Problem Statement:

The rapid expansion of AI-related employment has created inconsistent and overlapping job titles across industries, making it difficult for job seekers and analysts to understand how AI roles differ in compensation, experience expectations, work arrangements, and market positioning. This project aims to identify meaningful categories of AI jobs across North America and explain the characteristics that distinguish them and what can be expected of them.

By using data leveraged from multiple job postings related to Artificial Intelligence & technology jobs, I was able to tackle this task. The source dataset contained 51,932 postings and 22 variables. After initial cleaning and filtering by country labels, the analysis used 8,766 postings labeled as being in the United States or Canada

### Data Wrangling:

I started the data wrangling process with obtaining the AI Jobs Dataset 2026 through KaggleHub. It loaded both an organized dataset and a raw postings file, then used the organized dataset for subsequent analysis. Its 22 variables included job title, employer characteristics, location, country, work arrangement, salary range, experience level, education, skills, specialization, and applicant count.

Initial checks found no missing values in the organized dataset and no exact duplicate rows. These checks established completeness at the table level, but did not establish that every field was accurate or that different rows represented distinct underlying vacancies. Unique URLs and collection timestamps can prevent otherwise similar postings from appearing as exact duplicates.

Three industry categories containing one posting each were excluded, leaving 51,929 rows. Company-size labels were standardized by combining two nearly equivalent descriptions of the 1,000–5,000 employee range. Education descriptions were also consolidated. In particular, the ambiguous requirement “Master's or PhD” was assigned to a master's category, a simplification that should be considered when interpreting education features.

The country filter retained the United States, Canada, and Mexico, but no Mexican records were present. The resulting dataset contained 4,370 postings labeled United States and 4,396 labeled Canada. Therefore, the geographic coverage was limited to the two represented countries, subject to the accuracy of those labels.

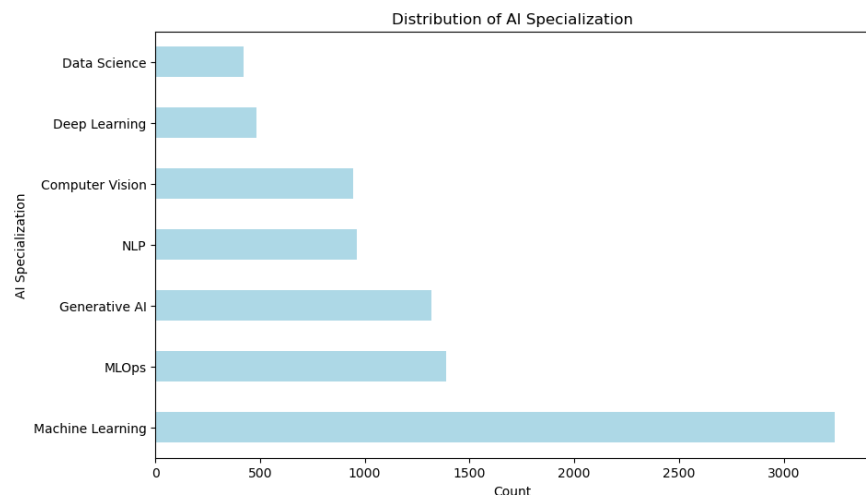
Salary ranges were parsed into lower and upper bounds. The prediction target was the arithmetic midpoint of those bounds. Although the exploratory notebook called this quantity “Median Salary,” it is a range midpoint, not the median of observed employee salaries. It also does not measure accepted offers or total compensation.

## Exploratory Data Analysis:

### Composition of the postings

Machine Learning accounted for the majority of the postings with 3,245 postings, followed by MLOps with ~1400 and Generative AI with ~1,300. NLP and Computer Vision contributed ~950 postings,

respectively, while Deep Learning and Data Science had ~500 and ~425. These figures describe the composition of this dataset and should not be interpreted as population estimates of demand across the AI labor market.

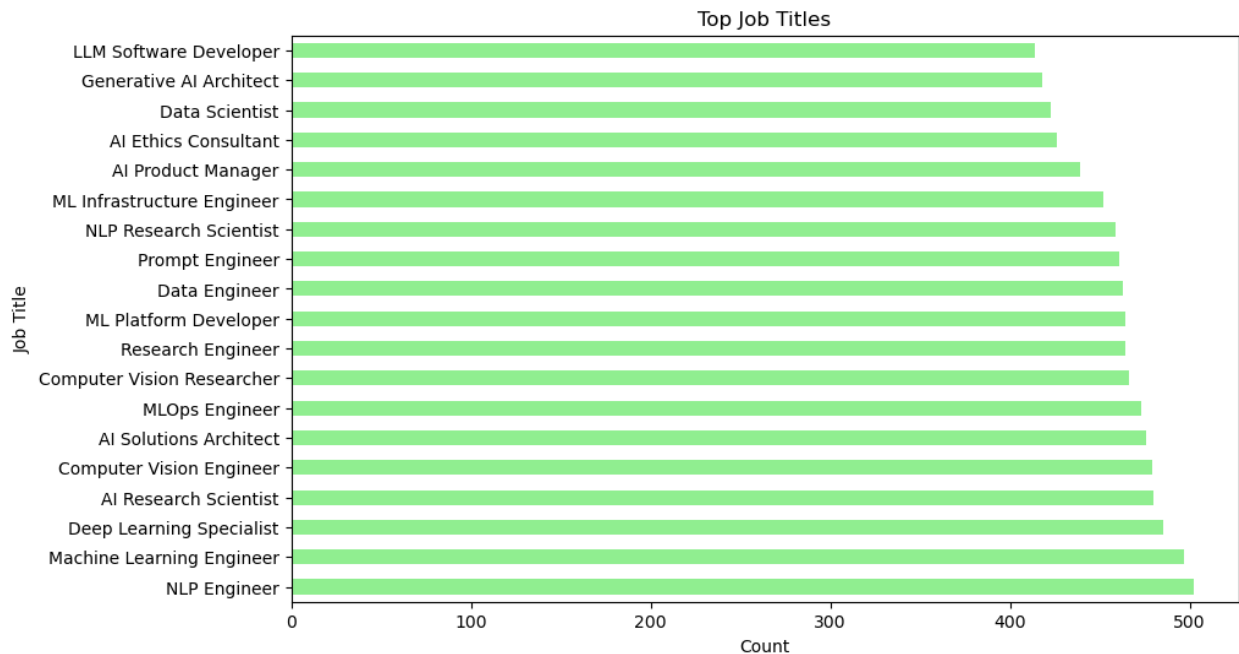


Mid-level positions represented ~4,500 postings, or approximately 50.2% of the sample. Senior roles accounted for ~2,500 postings and entry-level roles for ~1,725.

Remote positions were the largest work-arrangement group, with ~4,000 postings, followed by ~3,000 hybrid and ~1,800 on-site postings. Full-time employment accounted for ~7,500 postings, or approximately 85.2% of the sample. Counts across the ten retained industries were relatively similar.

## Job titles and technical skills

For detailed title and skill comparisons, the analysis focused on titles with more than 100 postings. This subset contained 19 titles and 8,741 rows, excluding 25 rows from the full country-filtered sample. The restriction made comparisons less sensitive to very small groups, but it also excluded uncommon titles that may still represent meaningful roles.



The recorded skill-frequency analysis identified Pandas as the most common skill, appearing in approximately 46.0% of this subset. Several traditional machine-learning tools and methods appeared in approximately 39–40% of postings, including LightGBM, Scikit-learn, Random Forest, Regression, XGBoost, NumPy, Supervised Learning, and Feature Engineering. PyTorch appeared in approximately 22.1%, while several MLOps tools appeared in approximately 19%.

Overall frequency did not capture how strongly skills distinguished individual specializations. The analysis therefore calculated skill prevalence within each specialization and subtracted prevalence across the full popular-title subset. The resulting values were percentage-point differences, not percentage increases.

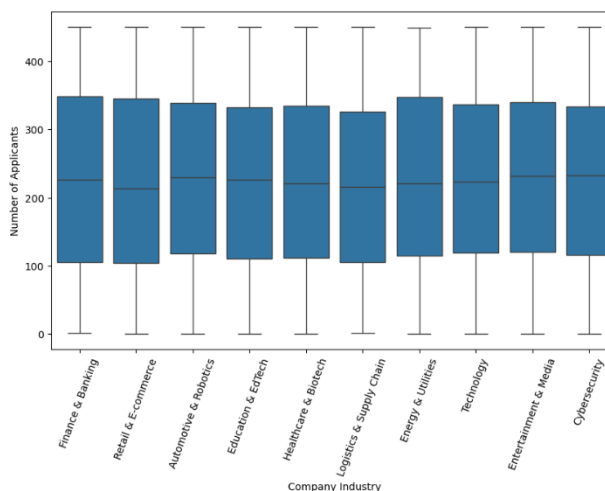
| Specialization          | Examples of distinguishing skills in the analysis                                    |
|-------------------------|--------------------------------------------------------------------------------------|
| <b>Computer Vision</b>  | OCR, ResNet, OpenCV, YOLO, image processing                                          |
| <b>Data Science</b>     | Matplotlib, Statsmodels, SQL, A/B testing, data visualization                        |
| <b>Deep Learning</b>    | Backpropagation, CNNs, CUDA, neural networks, Keras                                  |
| <b>Generative AI</b>    | Vector databases, LangChain, retrieval-augmented generation, fine-tuning, OpenAI API |
| <b>MLOps</b>            | Git, model monitoring, Kubeflow, Kubernetes, Terraform                               |
| <b>Machine Learning</b> | Feature engineering, supervised learning, NumPy, XGBoost, regression                 |
| <b>NLP</b>              | GPT, BERT, spaCy, tokenization, Transformers                                         |

These patterns support interpreting the dataset's specialization labels as distinct technical skill profiles. A skill's overall popularity and its usefulness for identifying a particular specialization are different properties. For example, a specialized tool can be uncommon across the full sample but highly characteristic of one group.

### Industry, work arrangement, and applicant volume

Within the popular-title subset, a chi-square test comparing industry and specialization produced a statistic of 61.93 and a p-value of 0.2141. A second test comparing specialization and work arrangement produced a statistic of 8.71 and a p-value of 0.7279. Neither test provided sufficient evidence to reject independence at a conventional 0.05 significance level. These results do not prove that the variables are unrelated.

Applicant counts showed substantial overlap across specializations, industries, work arrangements, and experience levels. Median counts by specialization ranged from 212 for Generative AI to 238 for Data Science, while within-group standard deviations were approximately 128–132 applicants. The descriptive results suggested that differences within groups were much larger than differences between their typical applicant counts. The analysis did not establish causal effects or build an applicant-volume prediction model.



## Preprocessing and Model Selection

### Models Selected

I chose to work with a Linear Regression model as it provided an interpretable starting point. I also worked with a Random Forest to try a different model that wasn't related to linear regression while wanting to see if I could find any nonlinear relationships and interactions by averaging predictions from multiple decision trees.

### Feature Engineering

The modeling dataset retained all 8,766 country-filtered postings. It contained 22 predictors: one normalized years-of-experience feature, three education indicators, three experience-level indicators, and fifteen binary skill indicators. Required experience ranged from zero to twelve years and was scaled to the interval from zero to one. Education was represented by bachelor's, master's, and PhD categories. Skill indicators used case-insensitive matching of complete comma-separated items, avoiding partial-name matches.

These feature choices defined the information available to the models. Hyperparameters, in contrast, controlled how each model learned from that information. For example, the number of selected skills was fixed during preprocessing, while tree depth was varied during Random Forest tuning. No additional scaling step was included in the model search. Linear Regression used 20 predictors after removing two reference-category indicators; Random Forest retained all 22.

### Training and Evaluation Approach

The data were divided into 7,012 training rows and 1,754 test rows using an 80/20 random split with a fixed seed of 42. Saved row identifiers kept predictors aligned with their salary targets. Five-fold shuffled cross-validation on the training set guided hyperparameter selection, and both model families used the same fold configuration.

Mean absolute error (MAE) was the primary evaluation metric because it expresses the average absolute prediction error in salary units. Root mean squared error (RMSE) provided a measure that gives greater weight to large errors.  $R^2$  measured performance relative to variation in the test targets. A mean-predicting DummyRegressor provided a simple reference for assessing whether the job features improved prediction.

## Modeling

## Linear Regression

Linear Regression provided an interpretable starting point. Bachelor's education and entry-level experience were omitted as reference categories, leaving 20 predictors. The same predictor columns were used for all four configurations. The search varied two settings:

| Hyperparameter       | Values Tested | What It Controls                                                                                                                    | Selected Value |
|----------------------|---------------|-------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>fit_intercept</b> | True, False   | Whether the model learns a constant starting value in addition to feature coefficients. With False, that constant is fixed at zero. | True           |
| <b>positive</b>      | False, True   | Whether coefficients can have either sign or must be nonnegative. This restriction applies to coefficients, not the intercept.      | True           |

Keeping an intercept allowed the model to represent a starting salary for the reference categories. Turning it off imposed a different constraint on the same feature matrix, and those configurations performed worse in validation. Allowing either sign for coefficients represented the original unconstrained model; restricting them to nonnegative values tested whether a simpler set of permitted relationships improved validation performance. This restriction does not guarantee positive salary predictions or establish that every feature increases salary.

The selected configuration was `fit_intercept=True` and `positive=True`. Its mean cross-validation MAE was \$9,735.49, with a fold standard deviation of \$157.62, compared with \$9,756.88 for the original settings (`fit_intercept=True`, `positive=False`). Tuning therefore reduced validation MAE by only \$21.39, approximately 0.22%. This improvement was small relative to the variation between folds. The selected model's mean training-fold MAE was \$9,719.57, close to its validation MAE, providing little indication of a large training-to-validation performance gap under this evaluation.

## Random Forest

Random Forest was used to capture nonlinear relationships and interactions by averaging predictions from multiple decision trees. Its search varied three hyperparameters, each with two candidate values, producing eight combinations:

| Hyperparameter          | Values Tested | What It Controls                                                                                                                                              | Selected Value |
|-------------------------|---------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>max_depth</b>        | True, False   | Maximum depth of each tree. None removes the depth limit, although other stopping rules still apply.                                                          | 8              |
| <b>min_samples_text</b> | 1,5           | Minimum number of training samples required in a leaf, the final group used to produce a tree prediction. Larger leaves discourage fitting very small groups. | 5              |
| <b>max_features</b>     | 0.5,1.0       | Fraction of predictors considered when searching for a split. The candidates correspond to half or all of the 22 features.                                    | 0.5            |
| <b>n_estimators</b>     | Fixed at 150  | Number of trees whose predictions are averaged. This value kept search time manageable.                                                                       | 150            |

The tree count was a fixed hyperparameter, so the experiment did not establish that 150 trees was optimal. The forest also used `random_state=42` for repeatable random sampling. Limiting depth and increasing leaf size restricted how closely individual trees could fit local details. Considering only half the predictors at each split introduced more variation among trees, which can reduce their tendency to make similar errors.

The selected configuration used `max_depth=8`, `min_samples_leaf=5`, and `max_features=0.5`. Its mean cross-validation MAE was \$9,559.79, with a standard deviation of \$110.46. Its mean training-fold MAE was \$9,258.59, giving a training-to-validation gap of about \$301. In contrast, forests with no depth limit and one-sample minimum leaves had training MAEs of approximately \$8,100 but validation MAEs of approximately \$10,100. Their closer fit to training rows did not translate into better validation predictions.

## Model Comparison

After tuning, each search refitted its selected model on the full training set. The following results compare those fitted models on the same held-out test rows; they are distinct from the validation scores used to select hyperparameters. The mean baseline was fitted using `strategy='mean'` and was not tuned.

| Model             | Test MAE    | Test RMSE   | Test R <sup>2</sup> |
|-------------------|-------------|-------------|---------------------|
| Mean Baseline     | \$40,567.00 | \$48,596.96 | -0.0016             |
| Linear Regression | \$9,478.15  | \$12,117.90 | 0.9377              |
| Random Forest     | \$9,250.28  | \$11,403.42 | 0.9449              |

## Model Interpretation

Experience was the dominant predictor group. In the Random Forest, normalized years of experience and the three experience-level indicators together accounted for approximately 96.8% of impurity-based feature importance. Experience-level indicators also had the largest Linear Regression coefficients. These results describe how the fitted models used the available features; they do not show that increasing experience causes a specific salary increase. Correlated predictors can share importance, and nonnegative coefficient constraints can force some linear coefficients to zero.

Overall, the Random Forest provided the best results.

## Future Work

The first priority is to correct currency conversion, validate country labels against locations, and regenerate all downstream salary results. The report's salary figures, model comparison table, and conclusions about model performance should then be updated together.

The predictive feature set omitted country, location, industry, work arrangement, title, and specialization. After data correction, testing these variables could establish whether they add information beyond experience. An experience-only model would also help quantify the incremental value of education and skills. Permutation importance and errors broken down by country, experience level, and specialization would provide a more complete assessment than aggregate metrics alone.

Finally, random splitting does not directly test performance on future postings or unfamiliar employers. A temporal split or employer-grouped evaluation would address those questions. Because the existing test results have already informed model comparison, further development should use a fresh final holdout or a nested evaluation design.

## Conclusion

The project provides a descriptive framework for interpreting AI jobs through specialization, skill requirements, and experience expectations. Its strongest descriptive finding is that existing specializations correspond to recognizable technical skill profiles. The saved models also show that experience-related features explain much of the variation in the currently processed salary targets, with Random Forest producing a small improvement over Linear Regression.

The work demonstrates an end-to-end workflow from data inspection and cleaning through feature engineering, tuning, and evaluation. Its salary conclusions remain provisional until currency conversion and geographic labels are corrected and the pipeline is rerun. Addressing those issues would make the final report more defensible and clarify how well its findings extend beyond this dataset.

As it is currently, work shown in this report could be used by HR representatives to figure out what skills to expect and look for in potential applicants, Executives looking to see what salary expectations to set, and by Academic Officials looking to determine what skills to teach to potential students for future courses.